

EXPERIMENT 4

Student Name: Shagun

UID: 24BDA70214

Branch: AIT-CSE

Section/Group: 24AIT_KRG-G2

Semester: 5

Subject Name: Computer Network

Subject Code: 24CSH-337

EXPERIMENT 4

Title: Study and Implementation of Static Routing using Cisco Packet Tracer

Aim

To create different network topologies using routers, switches, and PCs in Cisco Packet Tracer and configure **static routing** so that devices from different networks can communicate successfully.

Objectives

- To understand the concept of static routing.
- To create network topologies with routers, switches, and PCs.
- To assign IP addresses to all network devices.
- To manually configure static routes on routers.
- To test communication using the **Ping** command and **Simple PDU**.

Theory

Static Routing

Static routing is a method in which the network administrator manually adds routing information to every router. Each router is told where to send packets for other networks.

Unlike dynamic routing, static routing does not update automatically if the network changes. Therefore, if a new network is added or a link fails, the administrator must update the routing table manually.

Static routing is mostly used in **small and stable networks** because it is easy to configure, secure, and uses very little router memory and bandwidth.

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Crossover Cable

A crossover cable is used to directly connect two similar networking devices.

Examples:

- Router to Router
- Switch to Switch
- PC to PC

In this experiment, **Serial DCE/DTE cables (or crossover connection depending on topology)** are used to connect the routers so they can exchange routing information.

Header and Trailer

Whenever data is sent over a network, extra information is added.

Header

- Added before the actual data.
- Contains source IP address, destination IP address, protocol information, etc.

Trailer

- Added after the data.
- Contains error-checking information (CRC/FCS).

Advantages of Static Routing

- Easy to configure for small networks.
- Provides better security because routes are manually configured.
- Does not consume extra bandwidth.
- Uses less CPU and memory.
- Gives full control over packet forwarding.

Disadvantages of Static Routing

- Configuration must be done manually.
- Not suitable for large networks.
- Any network change requires manual updates.
- More chances of configuration errors.

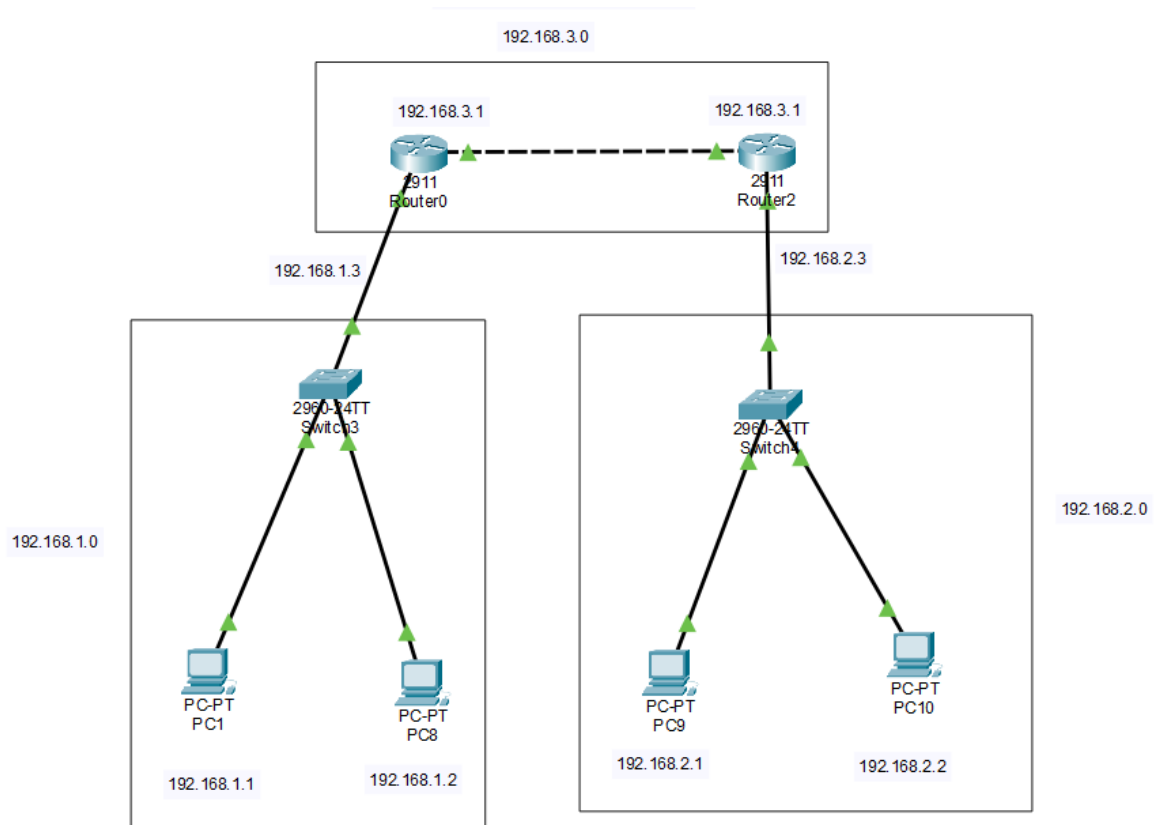
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Procedure

Part A – Network with 2 Routers, 2 Switches and 4 PCs

Steps

1. Open Cisco Packet Tracer.
2. Place **2 routers**, **2 switches**, and **4 PCs**.
3. Connect each switch to its router using a **Copper Straight-Through cable**.
4. Connect both routers using a **Serial DCE/DTE cable**.
5. Connect two PCs to each switch.
6. Assign IP addresses to all PCs and router interfaces.
7. Configure static routes on both routers.
8. Save the configuration.
9. Verify communication using **Ping** and **Simple PDU**.

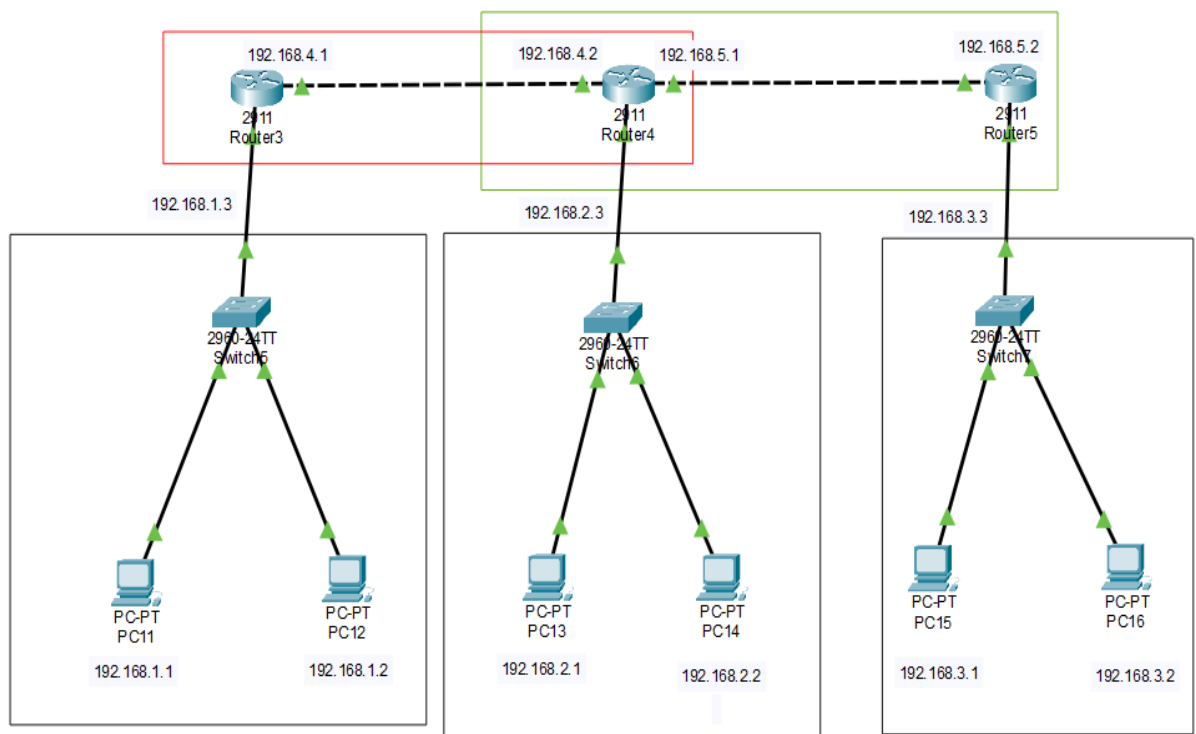


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Part B – Network with 3 Routers, 3 Switches and 6 PCs

Steps

1. Place **3 routers**, **3 switches**, and **6 PCs**.
2. Connect two PCs to every switch.
3. Connect each switch to its router.
4. Connect Router3 to Router4 and Router4 to Router5 using Serial cables.
5. Configure IP addresses for every PC and router interface.
6. Configure static routes on all three routers.
7. Check connectivity using **Ping**.
8. Test the complete network using **Simple PDU**.

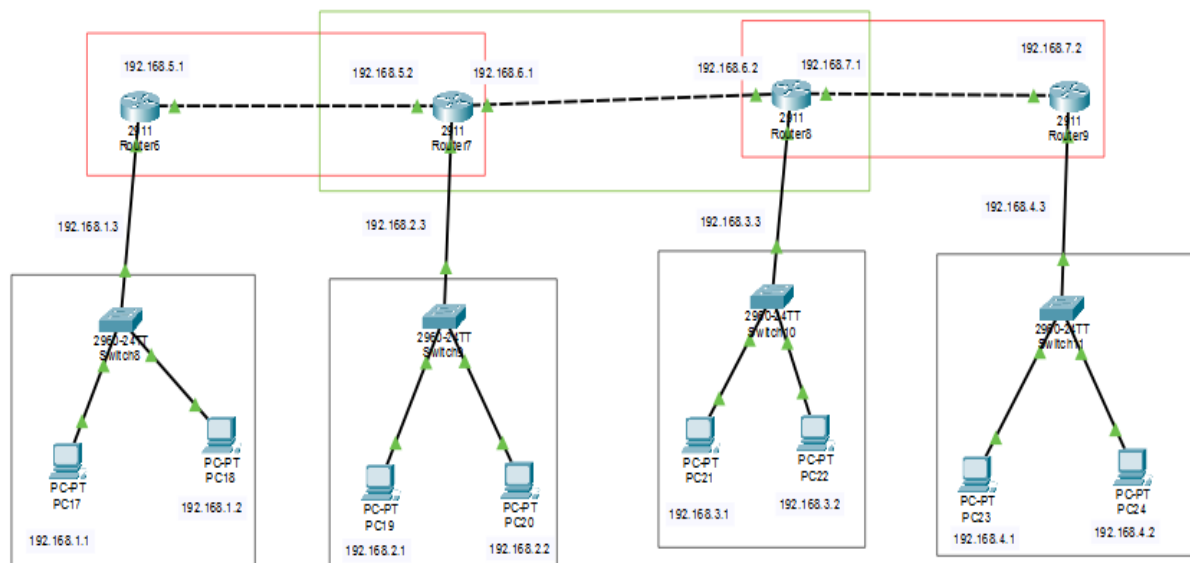


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Part C – Network with 4 Routers, 4 Switches and 8 PCs

Steps

1. Place **4 routers**, **4 switches**, and **8 PCs**.
2. Connect two PCs to each switch.
3. Connect every switch to its corresponding router.
4. Connect all routers in series using Serial cables.
5. Assign IP addresses to every PC and router interface.
6. Configure static routing on each router.
7. Save the configuration.
8. Use **Ping** and **Simple PDU** to verify that every PC can communicate with all other PCs.



Result

Static routing was successfully implemented in Cisco Packet Tracer for:

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- 2 Routers, 2 Switches, 4 PCs
- 3 Routers, 3 Switches, 6 PCs
- 4 Routers, 4 Switches, 8 PCs

All devices were able to communicate successfully after configuring the required static routes. Connectivity was verified using the **Ping** command and **Simple PDU**.

Learning Outcomes

After completing this experiment, I was able to:

1. Understand the working of static routing.
2. Build different LAN topologies in Cisco Packet Tracer.
3. Configure IP addresses and router interfaces.
4. Configure static routes manually.
5. Test communication using Ping and Simple PDU.
6. Understand how routing becomes more complex as the number of routers and networks increases.